

Vehicle Loan Default Prediction

A Credit Risk Model Using Real LTFS (L&T Financial Services) NBFC Data

A predictive model estimating the probability of default on vehicle loans, built on real historical lending data from LTFS, one of India's leading NBFCs. Compares an interpretable baseline (Logistic Regression) against a non-linear model (Random Forest), and identifies the key drivers of default risk.

Overview

Using 233,154 real vehicle loan records, this project predicts whether a borrower will default, and — more importantly for underwriting — explains *why*. The dataset reflects genuine challenges in Indian consumer lending: a 21.7% default rate (meaningful class imbalance) and over 50% of borrowers with no formal credit bureau history.

Key Findings

- **Loan-to-Value (LTV) ratio is the single strongest predictor of default** across both models. Default rates rise from 13.9% (LTV under 60%) to 25.9% (LTV 80–90%) — a clear, monotonic, and business-actionable relationship.
- **Random Forest modestly outperforms Logistic Regression** (AUC 0.626 vs. 0.613), suggesting the underlying relationships are largely linear, with only a small amount of additional non-linear signal.
- **50.2% of borrowers have no bureau history available** — a significant, real-world data quality issue handled explicitly (via a dedicated flag) rather than ignored or imputed carelessly.
- Model performance (AUC ~0.61–0.63) is consistent with published benchmarks on this dataset, reflecting genuine noise in real credit data rather than a modeling shortfall.

Methodology

1. **Feature engineering** — converted raw text/date fields (date of birth, “Xyrs Ymon” credit history strings) into usable numeric features; separated “no bureau score” from “the score value” to avoid the model misreading missing data as a literal zero score.
2. **Train/test split** — 80/20, stratified to preserve the true default rate in both sets.
3. **Two models trained and compared:**
 - Logistic Regression (with class balancing) — the industry-standard interpretable baseline for credit scoring
 - Random Forest — to check for non-linear feature interactions
4. **Evaluation** — AUC-ROC and precision/recall, not raw accuracy, since accuracy is misleading on this imbalanced dataset (a model predicting “no default” every time would score 78% accuracy while catching zero actual defaults).

Repo Contents

File	Description
<code>credit_risk_model.py</code>	Full pipeline — data loading, feature engineering, both models, evaluation, and all visualizations
<code>Credit_Risk_Model_Report.docx</code>	Full write-up: methodology, results, key drivers, business interpretation, limitations

Running the Model

1. Download `train.csv` from the [Kaggle dataset page](#) (requires a free Kaggle account)
2. Place `train.csv` in the same folder as `credit_risk_model.py` — or open the script and update the `DATA_PATH` variable near the top to point to wherever you saved it
3. Install dependencies and run:

```
pip install pandas numpy scikit-learn matplotlib
python credit_risk_model.py
```

This prints the full model comparison, classification report, and feature importance, and saves three charts (`roc_curve.png`, `feature_importance.png`, `default_by_ltv.png`) plus two CSVs of feature coefficients/importances.

Data Source

L&T Vehicle Loan Default Prediction dataset, originally released for the L&T Financial Services “Data Science FinHack” competition, via [Kaggle](#).

Limitations

- Class imbalance means metrics must be read as AUC/precision/recall, not accuracy
- Missing bureau history for ~50% of borrowers limits risk differentiation within that segment
- Baseline models only — no hyperparameter tuning or advanced feature engineering (interaction terms, etc.)
- Predicts default on the first EMI specifically, not lifetime default probability over the full loan tenure

See Section 8 of the full report for the complete list of assumptions and their justification.